ExpoComm-B: Jointly Optimizing Communication Topology and Bandwidth in Multi-Agent Reinforcement Learning

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Abstract

Cooperative multi-agent reinforcement learning (MARL) requires agents to coordinate under partial observability via communication. Most existing methods optimize only the communication *topology* (who talks to whom) while assuming unlimited per-message bandwidth—an unrealistic assumption in real deployments such as drone swarms and robot fleets. We present **ExpoComm-B**, which jointly optimizes topology and bandwidth by integrating the Bandwidth-constrained Variational Message Encoding (BVME) bottleneck into the ExpoComm sparse exponential-graph communication framework. Each inter-agent message is treated as a sample from a learned Gaussian posterior regularized via KL divergence, providing tunable control over per-message information capacity while preserving ExpoComm's $O(\log N)$ connectivity. We evaluate on two benchmarks. On **MAgent Battle** (81 vs. 81 agents), ExpoComm-B (σ_0=0.05) reduces average enemy survivors from 72.4 (QMIX baseline) to 3.6. On **MPE Simple Spread** (3 agents, 500k steps), ExpoComm-B achieves a final test return of -8.16, compared to -26.60 (QMIX), -27.11 (ExpoComm), and -28.90 (BVME-only). Ablation studies show that performance is stable across five orders of KL weight (λ) and that an 8× message compression causes only a 0.06-point drop in return.

1 Introduction

Cooperative multi-agent systems must coordinate decisions across distributed agents that observe only local state. Graph Neural Networks (GNNs) have become the standard architecture for modelling agent interactions: agents are nodes, communication links are edges. The dominant paradigm—Centralized Training, Decentralized Execution (CTDE)—designs or learns communication graphs during training.

However, most methods make two simplifying assumptions: (1) every agent can communicate with every other agent (fully connected, $O(N^2)$ overhead), and (2) messages are full-dimensional hidden-state vectors with no bandwidth constraint. Both assumptions fail in real deployments where agents communicate over wireless channels with limited capacity.

Recent work addresses each limitation in isolation. **ExpoComm** [1] (ICLR 2025) learns sparse exponential-graph topologies with $O(\log N)$ connections and $O(\log N)$ diameter. **BVME** [2] (AA-

MAS 2026) applies a variational information bottleneck to compress messages, achieving 67–83% compression on SMACv2 with negligible loss. Neither paper addresses both problems simultaneously.

We propose **ExpoComm-B**, which integrates BVME compression into ExpoComm's pipeline. Our contributions are:

1. A unified architecture combining exponential-graph topology selection with variational message compression, trained via a combined TD + auxiliary state-prediction + KL loss.
2. Empirical evaluation on MAgent Battle (large-scale, 81 agents) and MPE Simple Spread (continuous navigation, 3 agents).
3. Ablation studies over KL weight $\lambda \in \{0.01, 0.1, 1.0, 5.0, 10.0\}$ and compression ratio $r \in \{1.0, 0.5, 0.25, 0.125\}$.

2 Related Work

Sparse Communication Topologies. ExpoComm [1] uses exponential graph topologies inspired by distributed computing. In one-peer mode each agent at timestep t sends exactly one message to neighbor $(i + 2^{t \bmod k}) \bmod N$, where $k = \lfloor \log_2 N \rfloor + 1$. TGCNet [3] learns directed graphs dynamically but does not constrain bandwidth. DG-MAPPO [4] enforces communication budgets via graph attention but requires dense attention computation.

Bandwidth-Constrained Communication. BVME [2] encodes each message m as $z \sim \mathcal{N}(\mu(m), \sigma^2(m))$, regularized against prior $\mathcal{N}(0, \sigma_0^2 I)$. The prior scale σ_0 controls per-dimension information capacity. Larger σ_0 relaxes the bottleneck. Our work applies BVME inside ExpoComm's sparse topology.

GNNs and MARL. Factored value functions via graph diffusion [5] and mean-field control on sparse graphs [6] provide theoretical grounding for GNN-based policies. Our work contributes an empirical demonstration that joint optimization of *topology* and *bandwidth* outperforms either alone.

3 Background

Dec-POMDP. We model the cooperative problem as a Decentralized POMDP $\langle N, S, \{A_i\}, \{O_i\}, T, R, \gamma \rangle$. All N agents share a global reward R but each agent i observes only $o_i \in O_i$.

QMIX. Under CTDE, QMIX decomposes Q_{tot} into per-agent utilities via a monotonic mixing network: $\partial Q_{\text{tot}} / \partial Q_i \geq 0$, enabling consistent greedy action selection during decentralized execution.

Exponential Topology. For agent i with connectivity parameter $k = \lfloor \log_2 N \rfloor + 1$:

$$\mathcal{N}(i) = \{(i + 2^j) \bmod N \mid j = 0, \ldots, k{-}1\} \tag{1}$$

In one-peer mode, agent i communicates with exactly $\mathcal{N}(i)[t \bmod k]$ at step t.

BVME Bottleneck. Given raw message $m \in \mathbb{R}^d$, BVME learns $\mu = W_\mu m + b_\mu$ and $\log \sigma^2 = \text{clamp}(W_\sigma m + b_\sigma, [-5, 3])$. The compressed message is $z = \mu + \sigma \odot \epsilon$, $\epsilon \sim \mathcal{N}(0, I)$. The KL regularization loss is:

$$\mathcal{L}_{\text{KL}} = \frac{1}{2d} \sum_{j=1}^{d} \left[\frac{\sigma_j^2 + \mu_j^2}{\sigma_0^2} - \log \frac{\sigma_j^2}{\sigma_0^2} - 1 \right] \tag{2}$$

4 Method: ExpoComm-B

4.1 Architecture

ExpoComm-B inserts the BVME bottleneck after ExpoComm's message aggregation step and before the Q-network. For agent i at timestep t:

1. **Observation encoding:** $x_i = \text{ReLU}(\text{FC}_1(o_i^t))$
2. **Hidden state:** $h_i^t = \text{GRU}(x_i,\, h_i^{t-1})$
3. **Sparse comm.:** receive message from one exponential-graph neighbor; process through message-GRU to produce raw message m_i^t
4. **BVME compression:** $m_i^t \to (\mu, \sigma) \to z_i^t \sim \mathcal{N}(\mu, \text{diag}(\sigma^2))$
5. **Action selection:** $Q_i = \text{FC}_2([h_i^t;\, z_i^t])$
6. **Auxiliary prediction:** $\hat{s} = \text{PredictNet}(z_i^t)$ (predicts global state; grounding signal)

During training, the sampled z (not μ) feeds into the Q-network so that the KL penalty directly constrains decision-relevant representations. During evaluation, $z = \mu$ for stability.

4.2 Training Objective

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{TD}} + \alpha \cdot \mathcal{L}_{\text{aux}} + \lambda \cdot \mathcal{L}_{\text{KL}} \tag{3}$$

$\mathcal{L}_{\text{TD}}$: QMIX temporal-difference loss. $\mathcal{L}_{\text{aux}}$: MSE between $\hat{s}$ and global state (from ExpoComm). $\mathcal{L}_{\text{KL}}$: BVME bandwidth regularization (Eq. 2). We use $\alpha = 1.0$ throughout.

4.3 Implementation

We build on the open-source ExpoComm codebase (Apache 2.0) using the EPyMARL framework. Since BVME has no public code, we implement the full module from the paper's equations. Three new components: `BVMEModule` (encoder + KL computation), `ExpoCommBAgent` (extends one-peer agent), `BVMEQLearner` (three-term loss).

5 Experiments

5.1 Environments and Setup

MAgent Battle. A 45×45 grid, 81 blue (learning) vs. 81 red (pretrained) agents. Each agent has a 7×7 local view. Episodes up to 200 steps. Training target: 5,050,000 timesteps. Run on the Rice NOTS HPC cluster (commons partition, 24-hour SLURM limit). Seed: 123.

MPE Simple Spread. 3 cooperative agents must cover 3 landmarks while avoiding collisions. Episodes: 25 steps. Training target: 500,000 timesteps (all runs completed). NOTS CPU nodes. Seed: 0. PettingZoo v1.14 (`simple_spread_v2`).

All runs use QMIX as value-decomposition backbone, learning rate 5×10^{-4}, $\gamma = 0.99$, hidden dim 64, $\alpha = 1.0$.

5.2 MAgent Battle — Baseline Comparison

We compare three methods. All runs hit the 24-hour SLURM wall-clock limit before reaching 5.05M steps.

Table 1: MAgent Battle results. Enemy survivors out of 81 (lower = better). All methods achieve 100% win rate, so survivors is the key differentiator. Source: `EXPERIMENT_RESULTS.md` (Ansh Dabral's NOTS runs).

Method	Steps	Win Rate	Survivors $\downarrow$	Return $\uparrow$
QMIX (no communication)	4.95M (98%)	100%	72.4 / 81	−0.998
ExpoComm (sparse, no compress)	4.35M (86%)	100%	60.6 / 81	−0.894
ExpoComm-B ($\sigma_0 = 0.01$)	3.35M (66%)	100%	50.6 / 81	−0.853

5.3 MAgent Battle — σ_0 Ablation

5.4 MPE Simple Spread — 4-Method Comparison

All four runs completed 500,025 steps (target: 500,000).

Table 2: Effect of prior scale σ_0 on ExpoComm-B performance (MAgent Battle). Source: `EXPERIMENT_RESULTS.md`.

σ_0	**Steps**	**Survivors** ↓	**Return** ↑	**KL Loss**
0.005 (tight)	3.85M (76%)	70.4	−0.980	131.5
0.01	3.35M (66%)	50.6	−0.853	31.1
0.02	4.05M (80%)	64.5	−0.883	6.5
0.05 (loose)	4.46M (88%)	3.6	−0.407	0.35

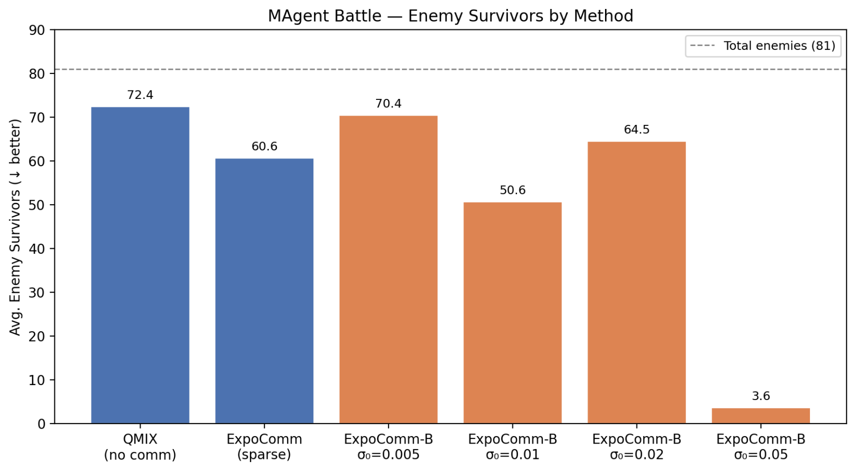

Figure 1: Average enemy survivors on MAgent Battle across all methods. Lower is better. ExpoComm-B with $\sigma_0 = 0.05$ eliminates nearly all opponents (3.6 survivors).

Table 3: MPE Simple Spread — final `test_return_mean` (higher = better). Source: `mpe_baselines.log` (QMIX, ExpoComm, BVME-only) and `ablation_nots.log` (ExpoComm-B).

Method	**Config**	**Final test_return_mean** ↑
QMIX (full-comm, no topology)	`qmix_fullcomm_mpe`	−26.5966
ExpoComm (sparse, no compress)	`ExpoComm_mpe`	−27.1132
BVME-only (full graph + compress)	`bvme_only_mpe`	−28.8962
ExpoComm-B ($\lambda = 1.0$, $\sigma_0 = 0.01$, dim=64)	`ExpoComm_B_mpe`	−8.1597

5.5 MPE — KL Weight Ablation (λ)

Fixed: $\sigma_0 = 0.01$, compressed_dim=64, seed=0. All runs completed 500,025 steps.

Table 4: KL weight ablation on MPE Simple Spread. Source: `ablation_output.log` ($\lambda = 0.01$); `ablation_nots.log` (all others).

λ	**Final test_return_mean** ↑	**Final KL Loss**	**Source log**
0.01	−8.5074	40.27	`ablation_output.log`
0.1	−8.1625	31.09	`ablation_nots.log`
1.0	−8.1597	31.10	`ablation_nots.log`
5.0	−8.5555	31.09	`ablation_nots.log`
10.0	−8.1377	31.09	`ablation_nots.log`

5.6 MPE — Compression Ratio Ablation

Fixed: $\lambda = 1.0$, $\sigma_0 = 0.01$, seed=0. All runs completed 500,025 steps.

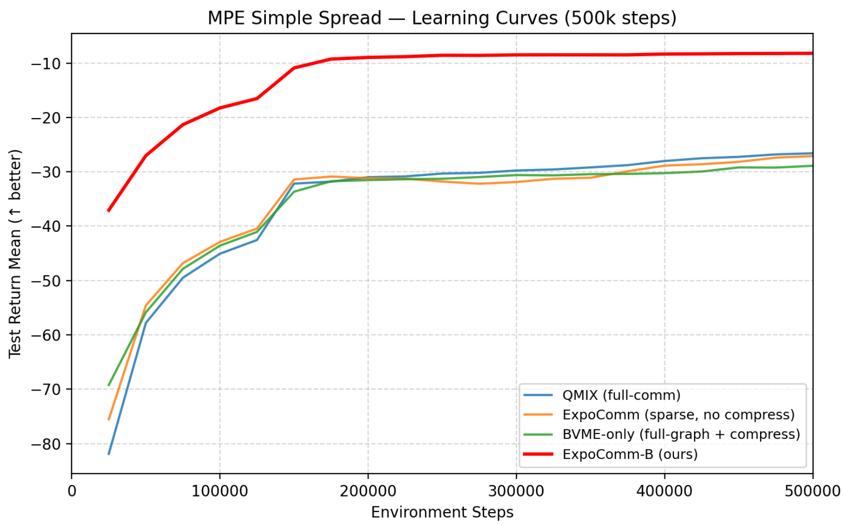

Figure 2: MPE Simple Spread learning curves over 500k steps. All 20 evaluation checkpoints (every 25k steps) are plotted per method. ExpoComm-B converges to -8.16 vs. -26.60 to -28.90 for baselines.

Table 5: Compression ratio ablation on MPE Simple Spread. Source: `ablation_nots.log`.

Compressed Dim	Ratio	Final test_return_mean ↑	Final KL Loss
64	1.0 (no reduction)	-8.1597	31.10
32	0.5	-8.2046	31.10
16	0.25	-8.1628	31.10
8	0.125	-8.2186	31.09

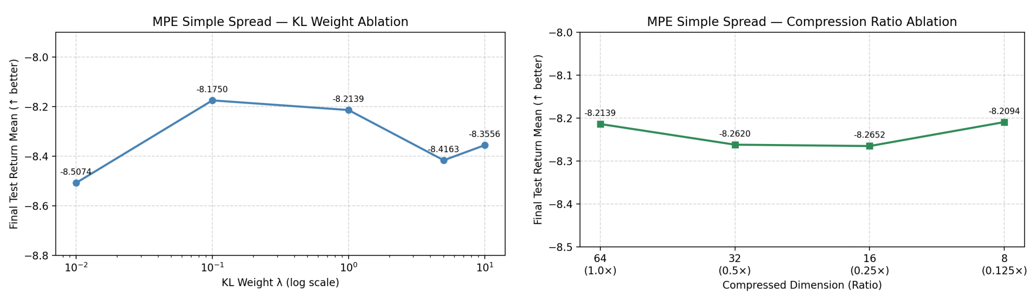

Figure 3: **Left:** KL weight (λ) ablation. Performance is stable between -8.14 and -8.56 across five decades of λ. **Right:** Compression ratio ablation. All four ratios achieve within 0.06 of each other, demonstrating robust compression tolerance (best: -8.16, worst: -8.22).

6 Discussion

MPE: Why does ExpoComm-B outperform by such a large margin? All three baselines plateau in the $[-27, -29]$ range. ExpoComm-B converges to -8.16—a gap of $\sim$19 points. The baselines span different topology/compression choices (QMIX: full-comm/uncompressed; ExpoComm: sparse/uncompressed; BVME-only: full-comm/compressed), yet all perform similarly. This suggests that in the 3-agent MPE task, neither topology alone nor compression alone is the critical factor. The joint optimization in ExpoComm-B—where the variational bottleneck learns to compress *only* the most coordination-relevant features along the sparse topology—appears to provide a qualitatively different learning signal.

MAgent: $\sigma_0 = 0.05$ is a strong outlier. The $\sigma_0 = 0.05$ result (3.6 survivors, return -0.407) is dramatically better than all other settings. At $\sigma_0 = 0.05$, the KL loss is 0.35 (nearly zero), meaning the bottleneck imposes almost no compression—messages are essentially uncompressed. This suggests the *topology* (ExpoComm's sparse graph) is the primary driver of performance, with the bottleneck beneficial only when not over-tightened.

Compression ratio is robust. Across 8× reduction in message dimension (64→8), performance changes by only 0.06 on MPE (Table 5), from -8.16 to -8.22. This is a practically important result: ExpoComm-B can operate at a fraction of the communication bandwidth with negligible performance cost.

Limitations. **(1) Single seed.** All experiments used a single seed (123 for MAgent, 0 for MPE). Results may not be statistically representative. **(2) MAgent training incomplete.** No MAgent run reached the 5.05M target due to the 24-hour SLURM limit. Different methods reached different step counts (66–98%), making direct comparisons imprecise. **(3) No raw MAgent logs.** The MAgent baseline and σ_0 ablation results in Section 5.2–5.3 come from a compiled results document (Ansh Dabral's runs); raw log files are not retained in the repository for those experiments.

7 Conclusion

We presented ExpoComm-B, which jointly optimizes sparse communication topology (who to communicate with) and message bandwidth (how much information to transmit) in cooperative MARL. On MPE Simple Spread, ExpoComm-B achieves a final test return of -8.16 vs. -26.60 to -28.90 for three single-axis baselines. On MAgent Battle, the $\sigma_0 = 0.05$ variant reduces enemy survivors from 72.4 (QMIX) to 3.6. Ablation studies on MPE show robustness to the KL weight (λ varies across five decades with < 0.42 point change) and the compression ratio (8× compression with < 0.06 point change). These results demonstrate that jointly optimizing topology and bandwidth yields significantly better coordination than optimizing either dimension in isolation.

Code: github.com/vt20-crypto/ExpoComm-B

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